

# Lia — Phase Roadmap

What's done, what's next, and what's deliberately not scheduled.

What's actually done, what's next, and what's deliberately not scheduled — kept here so a decision doesn't have to be re-derived from memory or git log next time. Rebuilt whenever a phase finishes; see the Growth Log for the narrative version of how she got here.

## Phase 1 — Polish what's built · done

Three small, independent items, chosen over widening her actions or chasing raw intelligence first. All three landed.

### 1.1 — Instrument the cloud fallback

So the free/paid/local model choice gets decided from a week of real data instead of one session's impression. Every rate-limit and every mid-stream drop now writes a timestamped line to `cloud_fallback_log.jsonl`, and `fallback_report.py` reads it back as a plain count.

*Already earning its keep: one real fallback landed within days of shipping this — a rate limit on the free conversation model, announced and absorbed rather than left silent.*

### 1.2 — Voice-ID status, spoken

`/whoami` was console-only, unreachable from the tray app most actual use happens in. “Do you know my voice”, “is that me” and similar now reach the same status out loud.

### 1.3 — Forget a single note

There was no way to remove one stored note short of touching the database directly. `db.delete_fact()` plus “what have you remembered” / “forget number 2” — numbered list, pick by position, the same pattern already proven for music.

## Asking someone else · done, not originally scoped

Added mid-stream, at request, rather than planned in the phase above: “ask cat for a suggestion” reaches a second, named free model and relays the answer attributed. Two names today — cat, fox — each measured and picked for being fast *and* reliable, not for novelty.

**Not the multi-agent idea this plan shelved below.** That idea split *her own* reasoning across several small local models. This is the opposite shape: one cloud call, on request, to a model she isn't and doesn't pretend to be, at zero local VRAM cost. Closer to the “look that up” internet path than to anything rejected.

Sixteen free OpenRouter models were catalogued directly and eight measured for real, twice. The metric mattered: this path collects the whole reply before speaking any of it, so total time is what a person waits on, not time to first word — the first pick (measured on the wrong metric) was heavier than it needed to be, and swapped once that was noticed. Two further names were tried and dropped rather than kept as a weak link: one lightest-on-paper replacement worked once and then failed empty five times in a row on a re-test; another was fast when it worked but wrong close to a third of the time across repeated real testing, each failure a real wait rather than an answer. Two names that have never once failed beat a bigger roster with a weak link in it.

*A real bug surfaced along the way, unrelated to model choice: a streamed request's timeout only bounds the gap between chunks, not the whole call, so a slow provider could hang far past it. Measured directly at 121 seconds, still empty. Every ask is now capped on wall-clock time regardless of provider behavior.*

## A typed panel and a new icon · done, not originally scoped

The tray app's first window. A small typed panel now opens alongside the tray icon, styled after Claude Code's own terminal — white and orange. Built as a second interface to the exact same conversation pipeline the console already had, not a new one: it satisfies the same typed-input and printed-output shapes `main.py` already knew how to use, so nothing about the conversation loop itself needed to change.

`LISTEN_ENABLED` was off while this was the current way of working with her — a config flag, not commented-out code, specifically so the tray's own **Listening** checkbox keeps working live, no restart, the moment voice is wanted again.

***That moment has arrived.** The panel existed to cover for her not being able to hear cheaply; she can now, on the NPU, so `LISTEN_ENABLED` is back on and `TRAINING_PANEL_ENABLED` is off. `panel.py` stays in the tree rather than being deleted — it costs nothing switched off, and it is the only way to reach her on a machine with no working microphone.*

*The tray icon changed too: a purple cat face, replacing the plain warm dot. Same two states (filled/listening, hollow/not) kept rather than reinvented; the mouth is the one new signal — open for speaking, closed for idle — doing the same job the old inner dot did, more legibly.*

**The one real risk was threading, and it was checked, not assumed.** Tkinter and the tray icon's own loop both want the thread that started them. Verified before writing any of it rather than after something froze: the tray library documents running off the main thread as unsafe in general, safe specifically on Windows — which is what this project has been throughout, so that's the trade taken.

# A machine with no graphics card · done, not originally scoped

She moved to a Core Ultra 7 255U — twelve cores, 16GB of LPDDR5X-8533, an integrated GPU, an idle NPU, and no discrete card at all. Every performance note written before this was measured on a machine with a 4GB card, so the question was not whether she still ran but which of those notes survived contact with the new hardware.

**The surprise was that memory bandwidth barely changed.** LPDDR5X-8533 gives roughly 136 GB/s, which is the same order as the 4GB card's VRAM. Token generation is bandwidth-bound rather than compute-bound, so generation held up at 21 tok/s — against speech's ~5 tok/s, her voice is the bottleneck and the model is not.

## Hearing moved to the NPU

The old design put Whisper on the CPU so *it wouldn't compete with Ollama for the 4GB of VRAM*. With no card that reasoning inverts exactly: hearing and thinking both want the CPU and now genuinely fight. But the chip has an NPU doing nothing, and speech recognition is the fixed-shape workload an NPU is actually good at.

|  | 2.3s clip | 5.4s clip |
|--|-----------|-----------|
|  | 0.55s     | 0.59s     |
|  | 0.42s     | 0.50s     |
|  | 0.15s     | 0.20s     |
|  | 0.11s     | 0.16s     |

*OpenVINO on the CPU is barely better than what it replaced, and worse on the long clip. Moving to OpenVINO is not the win; moving off the CPU is — the seconds it returns are seconds the CPU keeps for the language model.*

**One thing the NPU cannot do:** take Whisper's vocabulary hint. A prompt changes the decoder's input length and the NPU compiles to fixed shapes, so asking for one raises rather than degrading. Measured cost of losing it: none on the samples tested — “Anaya” came through correctly without it. WHISPER\_HINT\_ENABLED carries the detail, and CPU and iGPU both still accept it.

## The wait was reading, not writing

Time to first word was 7.68s, which looked like a slow model and was not. Generation was fine throughout at 21 tok/s; her 575-token system prompt was being *read* at about 110 tok/s. Ollama caches the prefix, so it is a once-per-session cost — turns two onward were already about a second.

*llm.warm\_up( ) was sending an empty message list, which loads the weights but leaves the prompt unread — so the existing prewarm was warming the cheap half. It now takes her real system message, and startup pays the cost while nobody is waiting. 7.46s to 2.35s on the first reply.*

## Two smaller things the move surfaced

- **The judge is off.** It cost a model call on every turn — 2.3–2.9s here against 0.47s on the old machine — to decide whether to spend 0.05s searching. It had also quietly got less accurate, since JUDGE\_MODEL follows MODEL\_CHAT and gemma2:2b scored 11/12 against llama3.2:3b's 35/36. Re-measuring the thing judge.py says cannot work — a score floor — found the populations now overlap by one question rather than throughout, so LIBRARY\_MIN\_SCORE went 0.45 to 0.52 and took the job: 15/16 at 45ms, against 11/12 at 2.5s. Measured on three short documents, not the novel the original overlap came from, so this gets worse as the library grows.
- **A missing Ollama hung startup for the full three minutes.** The wait exists because she beats Ollama to the login on autostart, which is worth keeping; waiting for something that was never installed is not. Now detected and reported in about two seconds — carefully, because a stale PATH after an install looks identical to a missing one and the naive check would abandon a perfectly good install.

## Phase 2 — A background summarizer · next

Of the three “raise the ceiling” ideas that came out of the intelligence conversation, this is the one with no architectural friction.

CLOUD\_HISTORY\_TURNS is a hard cutoff — anything older than four turns just falls off the edge of what she sends. A summarizer removes that cliff cheaply because it reuses a pattern that already exists twice: the diary and fact-extraction both run *off the hot path*, once, locally, at session rollover — not on every turn.

- A new SUMMARY\_PROMPT in config.py, same shape as DIARY\_PROMPT.
- A session\_summaries table in db.py.
- A call from end\_session(), alongside the diary and fact-extraction calls already there.
- A build\_summary\_context() injector — doesn't exist yet, since the diary has no equivalent — that folds the last summary into what CLOUD\_HISTORY\_TURNS sends.

*Zero added latency on a live turn, as long as it stays scoped to session rollover the same way the diary already is.*

### How to know it worked

Hold a session past four turns, end it, confirm a new row in session\_summaries. Start a fresh session and check — via CLOUD\_LOG\_USAGE's token count, or a debug print — that the summary is actually part of what gets sent, not just written and ignored.

## Phase 3 — Grounding check, log-only · optional, lower priority

The other “raise the ceiling” idea, with a real conflict worth knowing before starting: replies stream to speech sentence-by-sentence on purpose — speaking sentence one while the rest still generates is exactly the latency-hiding win already measured and built. A verifier that blocks speech until it checks the whole reply undoes that directly. A per-sentence check has its own problem: a lone sentence like “yes, exactly” can't be judged alone, and it's a call per sentence, not per turn.

**So: log-only first, if at all.** Check the completed reply against whatever memory or library context was already retrieved for that turn — no new retrieval needed, it's already materialized before the cloud call — *after* she's already spoken, and log a mismatch rather than gate on it. That's a real signal on how often the cloud model actually invents something, without touching the sentence streaming that already works. A blocking version, if it ever happens, is a separate decision made from what the log-only pass finds.

### How to know it worked

Trigger a reply that draws on retrieved context, confirm a log line records the check. Separately: time-to-first-word before and after, to confirm it's unchanged — this phase's entire point is adding the check without paying the latency its own design is built to avoid.

# Explicitly not doing right now

## Why not, for now

Only reachable through `AUTO_EXTRACT_FACTS`, which stays off by choice. Revisit only if that changes.

The old reason was no spare VRAM on a 4GB card. That card is gone, and the reason is now worse rather than better. The judge alone was measured at 2.3-2.9s a call. The role split it would add already exists anyway — `judge.py/intent` vs `another judge.py`. (Distinct from “Asking someone else” above.)

Not a current priority. Two real prerequisites whenever it is picked up: `intent.py`'s tools are deliberately parameterized for a specific voice-ID; voice-ID is the natural consent gate for it but is unenrolled and unvalidated against a real human voice today.

Single-machine, Windows-specific project today. Not raised as a priority.

## Open questions

- **Which model carries the actual conversation.** Reopened by the hardware move, and currently answered by default rather than by decision: no `OPENROUTER_API_KEY` is set on the new machine, so she is fully local on `gemma2:2b` — chosen over `llama3.2:3b` to keep the download small, and noticeably plainer for it. Generation measured 21 tok/s with headroom, so `llama3.2:3b` is very likely affordable here; that is the next thing to try, not a settled choice.
- **Whether the accuracy claims hold against a real microphone.** Every transcription measured so far has been of synthesised speech, which is cleaner than a room. In particular “losing the vocabulary hint costs nothing” is a conclusion drawn from clean audio and should not be trusted until someone has actually talked to her.
- **A third (or fourth) “ask” name.** Two were tried and dropped rather than kept as weak links. Worth revisiting if OpenRouter's free catalog turns up something both light and consistently reliable — re-measure on total time, the way cat and fox were, before adding one back.